

Do pretrained word embeddings improve the ability of Logical Tensor Networks (LTNs) to detect hyponymy relations?

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Abstract

We investigate whether pretrained word embeddings enhance Logical Tensor Networks’ (LTNs) ability to detect hypernymy relations. Comparing distributional (GloVe) and hierarchical (Poincaré) embeddings across baseline classifiers, MLPs, and LTNs, we find that GloVe embeddings enable near-perfect LTN performance despite exhibiting poor cosine similarity for hypernym pairs. Hierarchical embeddings perform below chance on classification tasks. While embedding dimensionality affects satisfaction of observed hypernym facts, global graph signals (Katz centrality) selectively reduce existential closure satisfaction independently of representational capacity. Our results reveal tensions between local entailment structure and global hierarchical constraints, suggesting the need for integrated semantic-structural representations.

1 Introduction

1.1 Hypernymy

Hypernymy is a semantic *is-a* or *type-of* relation between terms or concepts. As an interface between knowledge representation and hierarchical linguistic structure, it translates into logical statements with ISA as a truth-conditional declarative. Serving dual roles as knowledge base and taxonomical linguistic unit, hypernymy can be depicted as an acyclic graph resembling knowledge ontology structures.

1.2 Hypernymy and Neuro-Symbolic Reasoning

Binary truth assignment for hypernymy diminishes gains from graph organization, particularly since distance between terms—measured by binary conditioning over transitive closure—may not capture more distant links necessary for grounding relationships in larger datasets or corpora. Deploying Logical Tensor Networks (LTNs) for hypernymy

identification offers an affordance to lift such ontology beyond first-order logic formulas constrained to vertex pairs, using fuzzy logical connectors to include graph distance in truth-assignment evaluation.

1.3 Contributions

We discuss how propositional relations provide language and structure for logical statements, and how reasoning is constrained by distributional and structural representations, potentially providing enhanced basis for sound inferences. The assumed utility of neuro-symbolic methods rests on LTN use for knowledge-graph completion, though evidence for hypernymy research is missing. Beyond interpretability, differentiable logical connectors offer an alternative to probability-based hypernymy identification, mitigating issues from the phenomenon’s vague status in distributional lexical semantics. We revise representation learning’s role in hypernymy identification and discuss advantages of aligning LTNs with pretrained embeddings for inference over hypernymy relations.

2 Background

2.1 NLP and Linguistics

In NLP and linguistics, substitutability serves as a basic hypernymy test. In lexical semantics, hypernymy datasets rely on manually constructed resources and distributional characterization correspondence (Geffet Dagan 2004). WordNet (Fellbaum, 1998) offers the most comprehensive manually annotated hypernymy resource, with relations based on synonym sets (synsets) selected using substitutability, linking hypernymy to sentence truth values and lexical conceptual organization. The distributional inclusion hypothesis posits that distributional context scope differentiates hyponyms and hypernyms, with contextual features included in the superordinate category. Research

into distributional compositional semantics of lexical entailment (Kartsaklis Sadrzadeh, 2016), extending word co-occurrence as meaning-entailing substitutability from the lexical entailment hypothesis (Geffet Dagan 2004) and introducing low-dimensional feature permutations to capture hypernymy’s partial ordering, addresses the causal upward monotone context constraint of distributional methods (Kartsaklis Sadrzadeh, 2016; Vu Schwartz, 2018).

2.2 Representation learning

Vector-based approaches to hypernymy have been thoroughly explored for detection and extraction tasks. Encoding taxonomical structure with distributional contextual features proves particularly effective. Overcoming static embeddings’ lack of positional encoding—critical for hypernymy resolution—inspired solutions: learning hypernymy and non-hypernymy representations jointly and disjointly (Yu et al., 2015; Wang et al., 2019), reconstructing WordNet taxonomy by factorizing adjacency matrices (Saedi et al., 2018), leveraging WordNet path encoding (Cho et al., 2020), and taxonomy-enhanced contrastive or adversarial training (Wang et al., 2019). Leveraging taxonomical hierarchy alone inspired interest in ordered representation learning (Vendrov et al., 2016), particularly with non-Euclidean embedding spaces (Nickel Kiela, 2017; Muzellec Cuturi, 2018). Poincaré embeddings prove robust and efficient for isolated projection of taxonomical hierarchies (Nickel Kiela, 2017).

2.3 Logical Tensor Networks

Neuro-symbolic methods offer a data-agnostic alternative to mixed hierarchical and distributional embedding training. Logical Tensor Networks (LTNs), integrating trainable vector representations with differentiable logical operators, excel at modeling set-theoretic hypernymy properties. LTN applications in knowledge graph completion promise lifting first-order logic primitives—of limited availability in distributional solutions—and approaching hypernymy axiomatically as an n-ary relation constrained by irreflexivity, transitivity, and asymmetry. LTN’s native implementation of logical operators over real-valued vectors, with optimization over logical axioms as loss, introduces missing positional encoding, redefining meaning entailment as propositional rather than distributional.

3 Training techniques/models

We organize the training according to LTN design, using LogReg and MLP as baseline, adjusted to selection of pretrained embeddings to evaluate performance on hypernymy detection with three approaches of increasing expressiveness.

3.0.1 Baseline evaluation

As baseline, we evaluate pretrained embeddings with a baseline classifier against an MLP with similar parameters optimized with logical loss, distinguishing hypernymy learned directly through spatial similarity from that acquired through structural or taxonomical information, with the unsupervised baseline as a lower-bound estimate.

3.0.2 Task-specific architecture

3.0.3 LTN

Finally, we integrate embeddings into Logical Tensor Networks (LTNs), modeling hypernymy as a logical predicate with training maximizing truth values of axioms encoding transitivity, asymmetry, and irreflexivity. Unlike classifiers, LTNs incorporate symbolic knowledge, exploiting implicit or explicit hierarchical cues from embeddings, assessing whether neuro-symbolic reasoning yields gains beyond geometric similarity or shallow supervised learning.

3.1 Performance metrics

Performance was evaluated using accuracy (proportion of correct predictions) for straightforward discrimination assessment, and ROC-AUC to account for class imbalances and capture discriminative capacity across all thresholds. ROC-AUC measures the true positive–false positive trade-off as a threshold-agnostic ranking criterion for hypernym–hyponym pairs. To control for polysemy and partial set membership, we introduce Katz centrality (Katz, 1953) from WordNet embeddings (Saedi et al., 2018), which factorizes influence from immediate neighbors and all nodes, weighting closer ones more heavily, yielding an alternative boolean score for synset transitive closure. For LogReg and MLP baselines, we used standard classification metrics and row-wise comparisons mimicking binary and ternary predicates with Katz weight, evaluating via AUC differences. For LTN’s soft decision boundaries, we used soft truth values per axiom, predicate, and global formula satisfaction (SatAgg) as dependent variables with representation features (Katz boolean, embedding type, size,

data source) as regressors, reporting F-statistics for OLS regression (single variable and multivariate).

4 Experiments

4.1 Pre-trained embedding datasets

To isolate representational effects on hypernymy identification, we evaluate pretrained embeddings differing in lexical encoding, treating them as fixed inputs unless specified.

4.1.1 GloVe

For distributional baselines, we employed GloVe embeddings (Carlson et al., 2025) at d50, d100, d200, and d300 dimensions, and trained GloVe on text8 (Tevet et al., 2024) and enwiki9 (Mahoney, 2011). GloVe’s dense vectors from factorizing weighted log co-occurrence matrices optimize global corpus statistics, with custom contextual windows potentially capturing hypernymy from distributional patterns. Pretrained embeddings were frozen during training to test whether purely distributional semantics support hypernymy detection and whether model training compensates for lacking explicit taxonomic structure.

4.1.2 Poincaré

We adopted the adjacency matrix implementation from Saedi et al. (2018), using Katz-style resolvent (Katz, 1953) with PPMI for downweighting polysemous words and PCA reduction, yielding a (25, 25, 1) tensor with scalars for term pairs (Katz). This contrasts with arithmetic code vectors (Said, 2023) over WordNet’s noun root padded to maximal length (Codeword), and AS-TEAL (Wang et al., 2020)—taxonomy-enhanced embeddings (Mikolov et al., 2013) conditioned on input terms (TEAL). All embedding weights were frozen. We introduced Poincaré embeddings (Nickel Kiela, 2017) trained on the seminar dataset (Poincaré Default) and complete WordNet (Poincaré WN), comparing local graph and global hierarchical encoding by introducing Katz scalars into predicate grounding. LTN training followed the random embedding setup (MLP Hyp2) with two rounds: baseline (Codeword, Katz, TEAL, Poincaré WN 25) and Poincaré complete. All runs included binary and ternary predicates with Katz scalars (MLP Hyp3). OLS reports were organized by run and combined.

4.2 Tasks

Experiments are conducted on the seminar dataset (Default), mapping it to pre-trained embedding pair-

wise, with random vectors OVW, with negative pairs sampled from unrelated terms. used to evaluate the three training strategies from Section 2.4: cosine similarity, supervised classifiers and Logical Tensor Networks, with performance measured using accuracy and ROC-AUC (Section 2.5).

5 Results and Analysis

5.1 GloVe embedding results

As a benchmark, for all GloVe dimensions, average cosine similarity for hypernym pairs is lower than for non-hypernyms, with the gap increasing with dimensionality. This corroborates static embedding evaluations (Alkaabi et al., 2025) on the seminar dataset, showing co-occurrence data does not encode hypernymy relations, pointing toward compositional distributional entailment complexities (Baroni et al., 2012). Hypernyms are more distant than random pairs, indicating GloVe captures general semantic similarity rather than asymmetric "is-a" relations. Table 1 summarizes LR and MLP performance across embedding dimensions. LR achieves highest accuracy and AUC at 50 dimensions, decreasing for larger embeddings. MLP shows slightly lower accuracy but competitive ROC-AUC, suggesting non-linear interactions provide discriminative power. While pretrained GloVe does not explicitly encode hypernymy, both classifiers extract signal, though gains diminish with increasing dimensionality.

Dim	LR Acc	LR AUC	MLP Acc	MLP AUC
50d	0.9619	0.9821	0.9326	0.9751
100d	0.9238	0.9784	0.9062	0.9646
200d	0.9179	0.9760	0.9091	0.9657
300d	0.9238	0.9744	0.9208	0.9695

Table 1: Performance of Logistic Regression (LR) and MLP classifiers on hypernymy detection for different embedding dimensions. Accuracy (Acc) and ROC-AUC (AUC) are reported.

The Logical Tensor Network achieved near-perfect separation on the test set. Predicted hypernymy scores (Figure 2) show positive pairs above 0.98 and negative pairs below 0.005, with both accuracy and ROC-AUC reaching maximum values. This indicates LTN’s combination of semantic representations and logical constraints outperforms shallow classifiers trained on representation geometry alone.

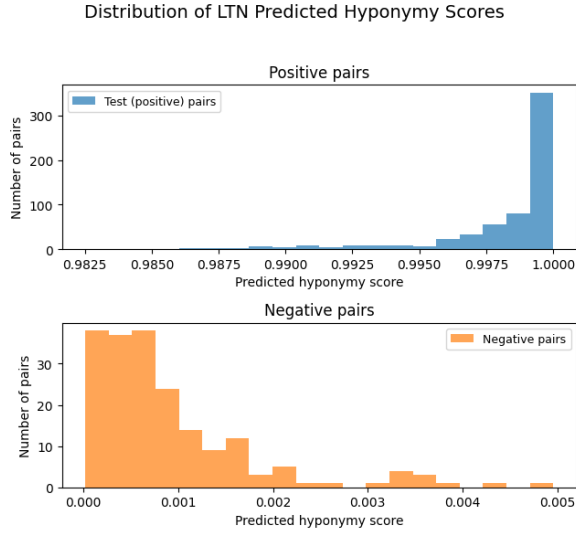


Figure 1: Distribution of predicted hyponymy scores for positive (test) and negative pairs using the LTN. Positive pairs (left) are all above 0.98, while negative pairs (right) are below 0.005.

5.2 Poincaré embedding results

Analysis of the baseline performance (see Table 2 in the Appendix) reveals an overall negative trend. Compared with hardcoded taxonomy (Codeword) and WordNet static embeddings with factorized centrality, hierarchical non-Euclidean embeddings (Poincaré) perform relatively poorly and mostly below chance level on the classification task. Since all representations used in the experiments have purely structural or taxonomical grounding, this suggests the need for cross-evaluation between Experiments 1 and 2 for complete falsification.

Already in the baseline analysis, a moderate negative Katz interaction is observed. Whereas the subsequent section explores this effect within a coherent embedding space, here the interaction spans across representation types, including otherwise highly scoring Codeword vectors.

In the LTN runs, logical axioms such as transitivity, irreflexivity, and asymmetry remain largely invariant across embedding choices once enforced within the LTN framework. In contrast, embedding dimensionality strongly affects satisfaction of observed hypernym facts, while global graph-based signals (Katz centrality) selectively reduce satisfaction of existential closure constraints. The negative effect of Katz on existential closure does not interact with embedding dimensionality, indicating that global centrality cues interfere with local entailment structure independently of representational capacity.

This tension is further evidenced by two observations: (1) semantically related hypernym terms are positioned further apart in the embedding space, and (2) when baseline labels are scaled to mirror similar conditions—specifically when target reach (accepted path depth) is proximal (3 steps)—Katz centrality also degrades baseline performance (MLP/LogReg).

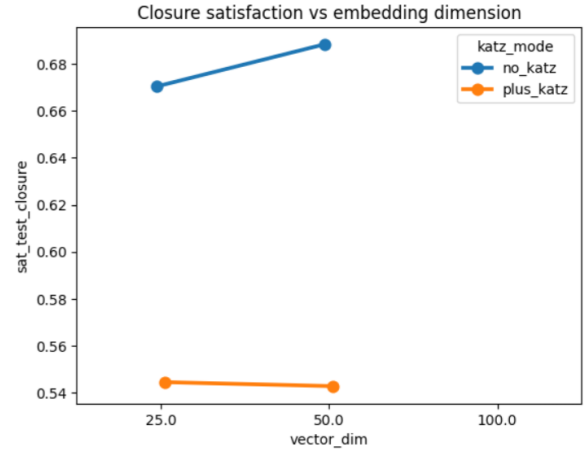


Figure 2: Interaction between embedding size and Katz weights over transitive closure satisfaction.

5.3 Comparing GloVe and Poincaré

The analysis presented here falls short of providing a conclusive answer to the question of the role of propositional relations as primitives for neuro-symbolic reasoning, nor does it clearly demonstrate how reasoning is constrained by distributional and structural representations.

Two contradicting tendencies emerged from Experiments 1 and 2: a moderate negative correlation between representation size and cosine similarity of positive matches on hypernymy pairs, and relative indifference to dimensionality when hierarchical representations are concerned. These findings would benefit from learning a projection from the Poincaré ball to Euclidean representation to explore potential synergies.

For LTNs, since the statistical embeddings experiment did not examine fine-grained axiom and predicate performance analysis, extending the scope of Experiment 1 would be a first step toward integrating these results. The tension between local and global constraints discovered in structural representations may be mirrored by increased distances between matching pairs in static counterparts, with the tension between co-occurrence and semantic entailment serving as an analogue.

6 Conclusions

Further work on this subject should focus on systematic exploration of the interaction between semantic and structural representations.

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Claim in text	Supporting evidence from OLS
Factorized centrality-driven embeddings perform poorly on baseline classification	Baseline OLS on <code>f1_macro</code> shows significant negative coefficients for <code>C(embed_type)[katz_scalar]</code> and <code>C(katz_mode)[plus_katz]</code> ; high R^2 (~ 0.93) despite small sample size.
Negative Katz interaction generalizes across representations	Katz coefficient remains significant even when Codeword vectors are used as the reference category; effect not limited to Poincaré embeddings.
Logical axioms invariant across embeddings in LTN	OLS models for <code>sat_train_trans</code> , <code>sat_train_irreflex</code> , and <code>sat_train_asymmetry</code> yield low R^2 and non-significant coefficients for <code>embed_type</code> , <code>vector_dim</code> , and <code>katz_mode</code> .
Embedding dimensionality affects observed hypernym facts	OLS for <code>sat_train_pos</code> shows a significant positive coefficient for <code>vector_dim</code> ($p \approx 0.02$).
Katz selectively reduces existential closure satisfaction	OLS for <code>sat_train_closure</code> exhibits a negative Katz coefficient with marginal significance; effect replicated in interaction models.
No Katz $\times$ dimension interaction	Interaction OLS shows <code>vector_dim:C(katz_mode)</code> non-significant, indicating independence of effects.
Effect mirrors baseline under depth-restricted labels	When accepted path length is restricted to ≤ 3 steps, Katz centrality degrades baseline MLP/LogReg performance in the same manner observed for existential closure.

Table 2: Summary of empirical claims and their statistical support from OLS analyses across baseline and LTN experiments.